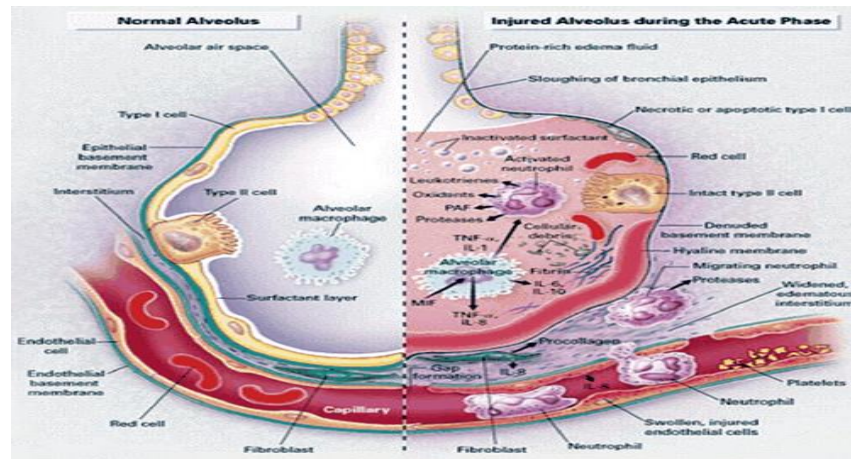


Acute Respiratory Distress Syndrom



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Introduction:

-Acute respiratory distress syndrome (ARDS):

is a life-threatening condition that causes lung swelling and fluid build up in the air sacs.

-The fluid buildup also makes the lungs heavy and stiff, and the lungs' ability to expand is severely decreased.

Definition:

Adult respiratory distress syndrome (ARDS), also called acute respiratory distress syndrome, is a type of lung (pulmonary) failure that may result from any disease that causes large amounts of fluid to collect in the lungs.

ARDS is not itself a specific disease, but a syndrome, a group of symptoms and signs that make up one of the most important forms of lung or respiratory failure

The basic fault is a breakdown of the barrier, or membrane, that normally keeps fluid from leaking out of the small blood vessels of the lung into the breathing sacs (the alveoli).

Common Causes of ARDS:

Direct lung injury (pulmonary ARDS):

1-Breathing in (aspiration) of the stomach contents when regurgitated, or salt water or fresh water from nearly drowning.

2-Massive trauma, with severe injury to any part of the body.

3-A blood clotting disorder called disseminated intravascular coagulation, in which blood clots form in vessels throughout the body, including the lungs.

4-A large amount of fat entering the circulation and traveling to the lungs, where it lodges in small blood vessels, injuring the cells lining the vessel walls.

Indirect injury (nonpulmonary ARDS):

- 1-Infection by a virus or bacterium, or [sepsis](#), a widespread infection that gets into the blood.
- 2-An overdose of a narcotic drug, a sedative, or, rarely, [aspirin](#) & alcohol overdose.
- 3- Inhaling smoke, as in a fire; toxic materials in the air, such as ammonia or hydrocarbons; or too much oxygen, which itself can injure the lungs.

4-Inflammation of the pancreas ([pancreatitis](#)), when blood proteins, called enzymes, pass to the lungs and injure lung cells

5-Severe burn injury.

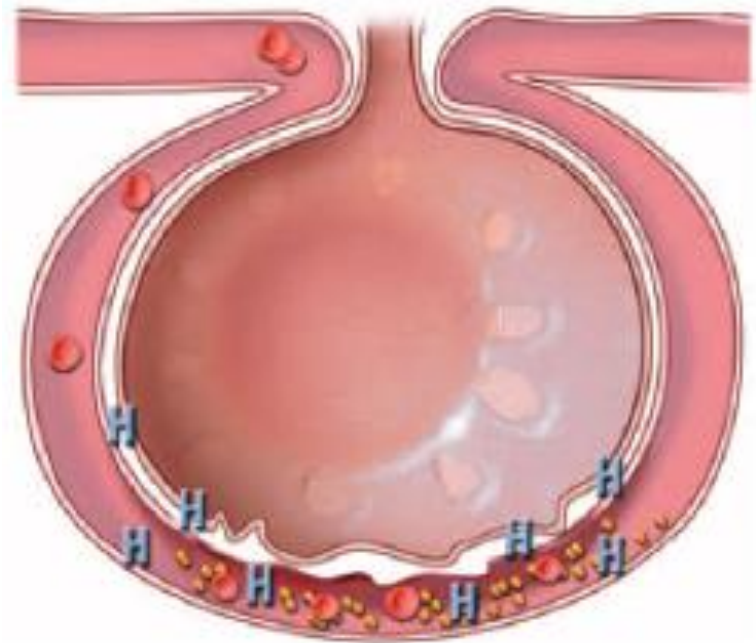
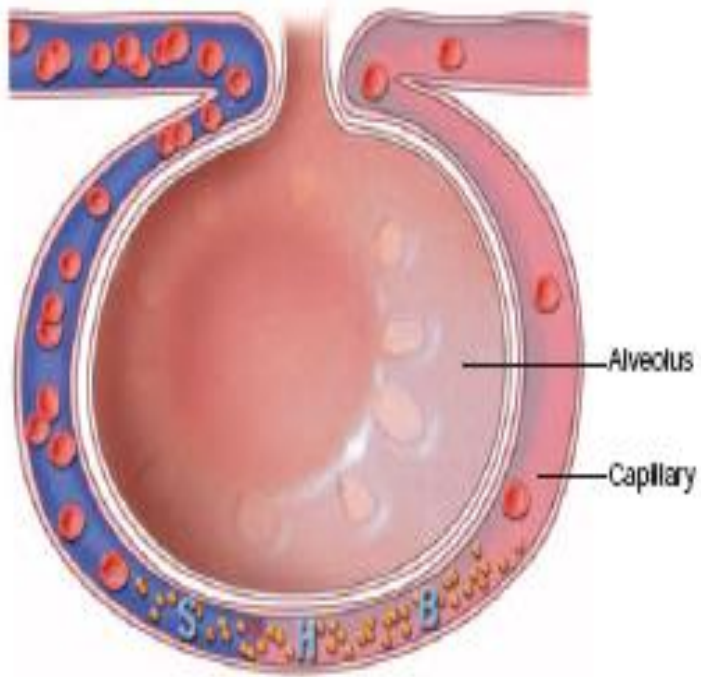
6- Shock with persistently low blood pressure may not in itself cause ARDS, but it can be an important factor.

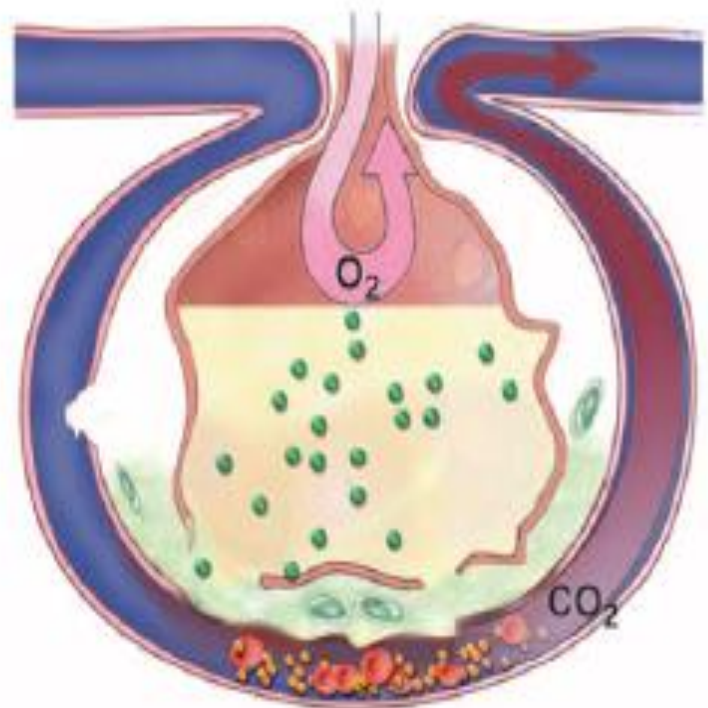
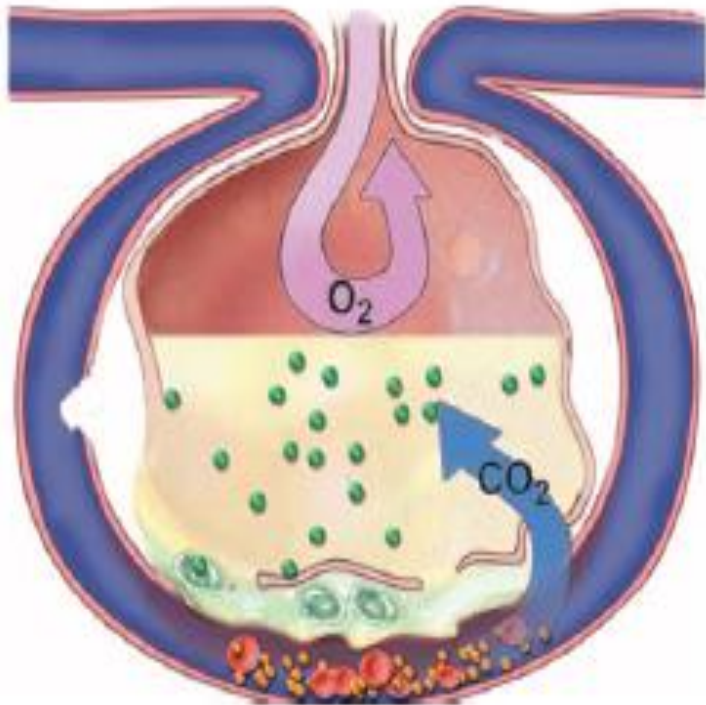
7-cardiopulmonary bypass.

PATHOPHYSIOLOGY:

ARDS is a clinical syndrome associated with a variety of pathological findings. the pathology most commonly associated with ARDS is diffuse alveolar damage(DAD).

DAD is characterized by a diffuse inflammation of lung parenchyma. The triggering insult to the parenchyma usually results in an initial release of cytokines and other inflammatory mediators, secreted by local epithelial and endothelial cells.





Diagnostic Test Results	Signs and symptoms	Stage& pathological change
• <i>ABG</i>: Respiratory alkalosis • <i>CXR</i>: No radiographic changes • <i>Chemistry</i>: Blood results may vary depending on precipitating cause (e.g., elevated white blood cell count, changes in hemoglobin) • <i>Hemodynamic</i>: Elevated PAP, normal or low PAWP	• Restlessness, dyspnea, Tachypnea • Moderate to extensive use of accessory respiratory muscles •	<i>Stage 1</i> (first 12 hr):No evidence of cellular damage.

- *ABG*: Decreased SaO₂ despite supplemental oxygen administration
- *CXR*: Patchy bilateral infiltrates
- *Chemistry*: Increasing acidosis (metabolic) depending on severity of onset
- *Hemodynamic*: Increasingly elevated PAP, normal or low PAWP

- Severe dyspnea, Tachypnea, cyanosis, tachycardia
- Coarse bilateral crackles
- Decreased air entry to dependent lung fields
- Increased agitation and restlessness

Stage 2 (24 hr)
Exudative stage:
neutrophil infiltration
,vascular congestion & interstitial edema.

- *ABG*: Worsening hypoxemia
- *CXR*: Air bronchograms, decreased lung volumes
- *Chemistry*: Signs of other organ involvement: decreased platelets and hemoglobin, increased white blood cell count, abnormal clotting factors
- *Hemodynamic*: Unchanged or becoming increasingly worse

- Decreased air entry bilaterally
- Impaired responsiveness (may be related to sedation necessary to maintain mechanical ventilation)
- Decreased gut motility
- Generalized edema
- Poor skin integrity and breakdown

Stage 3 (2–10 days)
proliferative stage:
cell proliferation & increase interstitial and alveolar edema.

- *ABG*: Worsening hypoxemia and hypercapnia
- *CXR*: Air bronchograms, pneumothoraces
- *Chemistry*: Persistent signs of other organ involvement: decreased platelets and hemoglobin, increased white blood cell count, abnormal clotting factors
- *Hemodynamic*: Unchanged or becoming increasingly worse

- Symptoms of MODS, including decreased urine output, poor gastric motility, symptoms of impaired coagulation
OR
- Single-system involvement of the respiratory system with gradual improvement over time

Stage 4 (>10 days) Fibrotic stage: Thickening of interstitial wall with fibrosis & cyst formation.

Tests used in the diagnosis of ARDS include:

1- The [chest x ray](#): may be normal in the early stages, but, in a short time, fluid will be seen where it does not belong. The two lungs are about equally affected.

2- A heart of normal size indicates that the problem actually is ARDS and not heart failure by [echocardiogram](#) (heart ultrasound).

3-Arterial blood gas (ABGs): Characterized by refractory hypoxemia (decrease P_{aO_2} unresponsive to increase F_{iO_2}). Initially $pH > 7.45$ because of hyperventilation. As ARDS worsens, pH falls to < 7.35 because of respiratory acidosis.

4-CBC and blood chemistries

5-Evaluation for possible infections

6-Cultures and analysis of sputum
specimens

Treatment:

1-General

Acute respiratory distress syndrome is usually treated with mechanical ventilation in the Intensive Care Unit.

Ventilation is usually delivered through oro-tracheal intubation, or tracheostomy whenever prolonged ventilation (≥ 2 weeks) is deemed inevitable

Mechanical ventilation:

Use PEEP to decrease intrapulmonary shunt.

Lower tidal volume to avoid lung over distension.

Accept hypercapnia.

Minimize FiO₂ to decrease risk of oxygen toxicity

PEEP in ARDS:

Improves oxygenation by reducing shunt.

May protect against ventilator induced lung injury.

The high pressures and other breathing machine settings required to treat ARDS often require that the patient be deeply sedated with medications.

Prone position

Distribution of lung infiltrates in acute respiratory distress syndrome is non-uniform. Repositioning into the prone position (face down) might improve oxygenation by relieving atelectasis and improving perfusion.

Treatment of the underlying cause is imperative, as it tends to maintain the ARDS picture.

Appropriate [antibiotic](#) therapy must be administered as soon as [microbiological culture](#) results are available. [Empirical](#) therapy *may* be appropriate if local microbiological surveillance is efficient. More than 60% ARDS patients experience a ([nosocomial](#)) pulmonary infection either before or after the onset of lung injury.

Fluid management

Several studies have shown that pulmonary function and outcome are better in patients lost weight or pulmonary wedge pressure that was lowered by diuresis or fluid restriction.

Corticosteroids

- study has found significant improvement in ARDS using modest doses of corticosteroids.
- The initial regimen consists of [methylprednisolone](#) 2 mg/kg daily. After 3-5 days a response must be apparent
- In 1-2 weeks the dose can be tapered to methylprednisolone 0.5-1.0 mg daily.
- Patients with ARDS do not benefit from high-dose corticosteroids.

Nitric oxide

Inhaled nitric oxide (NO) potentially acts as selective pulmonary vasodilator. Rapid binding to hemoglobin prevents systemic effects. It should increase perfusion of better ventilated areas

Complications:

Pulmonary: pulmonary embolism (PE), •
pulmonary fibrosis, ventilator-associated
pneumonia (VAP).

Gastrointestinal: hemorrhage (ulcer), •
dysmotility, pneumoperitoneum, bacterial
translocation.

Cardiac: arrhythmias, myocardial dysfunction. •

Renal: [acute renal failure](#) (ARF), positive fluid •
balance.

Mechanical: vascular injury, pneumothorax •
(by placing pulmonary artery catheter),
tracheal injury/stenosis (result of intubation
and/or irritation by endotracheal tube).

Nutritional: malnutrition (catabolic state), •
electrolyte deficiency

Many people who survive ARDS suffer [memory loss](#) or other problems with thinking after they recover. This is related to brain damage caused by reduced access to oxygen while the lungs were malfunctioning

Collaborative care for the Patient with Acute Respiratory Distress Syndrome (ARDS):

1- Oxygenation/Ventilation:

- 1-Auscultate breath sounds every 2 to 4 hours and as required.
- 2-Intubate to maintain oxygenation and ventilation and decrease work of breathing.
- 3-Suction endotracheal airway when appropriate
- 4-Hyperoxygenate and hyperventilate before and after each suction pass.
- 5-Monitor airway pressures every 1 to 2 hours.
- 6-Monitor airway pressures after suctioning.

7-Administer bronchodilators and mucolytics.

8-Obtain a PEEP study to determine optimal oxygen delivery.

9-Consider a change in ventilator mode to prevent barotrauma and volutrauma.

10-Turn side-to-side every 2 hours.

11-Perform chest physiotherapy every 4 hours, if tolerated.

12-Consider prone positioning.

13-Take chest x-ray daily.

2- Circulation/Perfusion

- Assess hemodynamic effects of initiation of mechanical ventilation (e.g., decreased venous return and cardiac output).
- Monitor electrocardiogram (ECG) for dysrhythmias related to hypoxemia.
- Assess hemodynamic effects of changes in inspiratory pressure settings, tidal volume, PEEP, and ventilatory modes.
- Assess effects of ventilator setting changes on cardiac output and oxygen delivery.
- Administer intravascular volume to maintain preload.
- Monitor vital signs every 1 to 2 hours.

3- Fluids/Electrolytes

- Monitor hydration status to reduce viscosity of lung secretions.
- Monitor intake and output
-
- Avoid use of nephrotoxic substances and overuse of diuretics.
- Administer fluids and diuretics to maintain intravascular volume and renal function
- Replace electrolytes as ordered.
- Monitor blood urea nitrogen (BUN), creatinine, serum osmolarity, and urine electrolytes as required.

4- Mobility/Safety

Initiate deep venous thrombosis (DVT) prophylaxis.

Reposition frequently.

Mobilize to chair when acute phase is past, hemodynamic stability and hemostasis achieved.

Consult physiotherapist.

Conduct range-of-motion and strengthening exercises when able.

Monitor mechanical ventilator alarms and settings and patient parameters (e.g., tidal volume) every 1 to 2 hours.

5- Skin Integrity

Assess skin every 4 hours and each time patient is repositioned.

Turn every 2 hours.

Consider pressure relief/reduction mattress, kinetic therapy bed, or prone positioning.

Use Braden Scale to assess risk of skin breakdown.

6- Nutrition

- Provide enteral nutrition within 24 hours.
- Consult dietician or nutritional support service.
- Monitor lipid intake.
- Monitor albumin, prealbumin, transferrin, cholesterol, triglycerides, and glucose.

Comfort/Pain Control

- Objectively assess comfort/pain using a pain scale.
- Provide analgesia and sedation as indicated by assessment.
- Monitor patient cardiopulmonary and pain response to medication.
- Provide continuous or routine (every 1 to 2 hours) intravenous sedation and analgesia.

8- Psychosocial care

- Assess vital signs during treatments, discussions, and the like.
- Cautiously administer sedatives.
- Consult social services, clergy, as appropriate.
- Provide for adequate rest and sleep.

any question

Thank You